

Notes, Comments, and Letters to the Editor

The Effect on Optimal Consumption of Increased Uncertainty in Labor Income in the Multiperiod Case*

We consider a multiperiod, additive utility, optimal consumption model with a riskless investment and a stochastic labor income. The main result is that for utility functions belonging to the set F , consumption decreases when we go from any sequence of distribution functions representing labor income to a more risky sequence. A concave utility function belongs to F if its first derivative exists everywhere and is convex.

I. INTRODUCTION

The impact on consumption of increased uncertainty in future labor or capital income has been examined by a number of authors in the last ten years. As illustrated by Sandmo [18], the answers one gets are different in the two cases of uncertain labor income and uncertain capital income. Therefore, in order to separate these effects, the models with random labor income generally have one nonrisky investment opportunity, and those with random investment opportunities have a deterministic (or zero) labor income. The model in this paper conforms to the above dichotomy. The only exception to that rule seems to be in Merton [11, Sect. 8], which treats the case of a nondecreasing Poisson income stream, an exponential utility function, and two investment opportunities, one riskless and the other described by Brownian motion.

Three relatively early papers which examine the random labor income case are the two-period models of Leland [8], Sandmo [18], and Drèze and Modigliani [5]. Their problem is: given the first period labor income y_1 and the distribution function Y_2 of the labor income in period two, choose consumption c_1 in period 1, $0 \leq c_1 \leq y_1$, so as to maximize $EU(c_1, (1+r)(y_1 - c_1) + Y_2)$.

Assuming the utilities are additive, ($U = u_1 + u_2$), Leland [8, Eq. (25)] concludes that concavity and a positive third derivative imply that there

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is a decrease in consumption when going from the deterministic income case to the random income case with the same mean and an infinitesimal random element (such that a second-order Taylor approximation is valid). Sandmo compares parameterized versions of Y_2 of the form $\alpha Y_2 + (1 - \alpha) E(Y_2)$, $0 \leq \alpha \leq k$, where k is such that the income remains nonnegative, and he demonstrates that c_1 is a decreasing function of α (and of risk) when U has decreasing temporal risk aversion. His results imply that in the case of a concave additive utility function, c_1 is a decreasing function of α when the third derivative is positive. In [5] Drèze and Modigliani look at the income and substitution effects of increased risk in labor income.

The model that we will be working with is an infinite horizon additive utility model which the author used in [12]. There the main qualitative result was that for isoelastic utility functions, consumption decreases when we go from the deterministic labor income case to the random labor income case with the same mean. Here we will show that for utility functions belonging to the set F , consumption decreases when we go from any sequence of distribution functions representing labor income to a more risky sequence where we are using increased risk in the sense of Rothschild and Stiglitz [15, 17]. A concave utility function belongs to F if its first derivative exists everywhere and is convex. Therefore if a concave utility function is thrice differentiable, then it belongs to F if its third derivative is nonnegative. It is an easy exercise to verify that the isoelastic utility functions, $(1/\gamma) c^\gamma$, $\gamma < 1$, $\gamma \neq 0$, belong to F .

In view of the importance of a nonnegative third derivative as exhibited in Leland [8] and Mirman [13], it is not surprising that the third derivative is also the key condition in this model. Its import is made all the more plausible when we recall the certainty-equivalence results of Theil [22], Simon [21], and recently Duchan [6]. Essentially, their results state that with a quadratic utility function (third derivative zero) and linear state equation with an additive random disturbance, the decisions are unaltered if the random elements are replaced by their means.

In the random capital income case with an isoelastic utility function, the effect on consumption of increased risk in the return on capital is different depending on whether $\gamma < 0$ or $\gamma > 0$ (Rothschild and Stiglitz [16]). Therefore we get the differing conclusions in the random labor income and random capital income models with an infinite horizon that Sandmo observed in his two-period model. In [4, p. 354], Diamond and Stiglitz have further analyzed and clarified the effect of increased risk in the random capital income case with an isoelastic utility function using the concept of mean utility preserving increase in risk.

We will also establish two secondary results. The first is Theorem 1,

which establishes an equivalent definition for F . The second is at the end of the paper where it is shown that the individual's expected utility decreases when we go from any sequence of distribution functions representing labor income to a more risky sequence.

Since completing this research, two papers, Schectman [19] and Sibley [20], whose results overlap mine, have been brought to my attention. Both papers contain other topics as well. Sibley obtains the same main result when the (concave) utility function has a nonnegative third derivative and an infinite first derivative at zero. This is a more restrictive class and, for example, the concave quadratic utility functions are included in our results but not his. Schectman works with same class of utility functions F as we do. He also proves our Lemma 1 (his Theorem 3.6) using a quite different method of proof. His main result is weaker since he compares the stochastic case with the deterministic case and does not use the concept of increasing risk.

2. THE MODEL AND THE MAIN RESULTS

Except for a more general class of utility functions, the model we consider is the same as that presented by Miller in [12], so that we will limit ourselves to the bare essentials and refer the reader to [12] for discussion of the model. Consider

(x, j) : the state of the system where x represents the capital at the beginning of period j .

$r - 1$: the rate of interest for both lending and borrowing where $r > 1$.

Y_j : the nonnegative random income received at the end of period j . It is convenient to divide Y_j into certain and uncertain parts by $Y_j = y_j + R_j$ where $y_j = \sup\{h: F_j(h) = 0\}$ and $F_j(\cdot)$ is the distribution function of Y_j . We also assume that the Y_j are independent, but not identically distributed, that the means of R_j are uniformly bounded, and that $\sum_{i=1}^{\infty} r^{-i} y_i < \infty$. It is significant that we do not assume that the Y_j are identically distributed, for otherwise the optimal decision in period j would depend on the value of x_j and not on j .

D_j : the amount of debt allowed in period j equals $\sum_{i=1}^{\infty} r^{-i} y_{j+i-1}$. D_j is finite by our assumption above concerning the y_j . Thus we allow the individual to borrow against certain future income and x_j can take on value in $[-D_j, \infty)$.

c_j : the consumption in period j . We require that $0 \leq c_j \leq x_j + D_j$.

$U(c_1, c_2, c_3, \dots)$: the utility function for all feasible $c_1, c_2, \dots$, equals $\sum_{i=1}^{\infty} \alpha^{i-1} u(c_i)$, where α is a discount factor between 0 and 1. We will restrict our attention to $u \in F$ where F is defined below.

DEFINITION. A concave function $g: I \rightarrow R$, with the convex set $I \subset R$, belongs to F if its first derivative, g' , exists everywhere on the interior of I and is convex.

In this paper I will be $[0, \infty)$ or $(0, \infty)$ if g is a utility function, and $[-D_j, \infty)$ if g is related to the optimal return function of period j . This class of utility functions was developed independently in a paper by Schechtman [19].

The decision making takes place as follows. In period 1 the individual has x_1 units of capital. He decides to consume c_1 , where $0 \leq c_1 \leq x_1 + D_1$, and he receives a utility $u(c_1)$. The resulting capital (or debt) grows to $r(x_1 - c_1)$ and a random income Y_1 is received at the end of period 1 so that x_2 equals $r(x_1 - c_1) + Y_1$. In general starting from state (x, j) the new state is given by

$$T(x, j) = (r(x - c) + Y_j, j + 1). \quad (1)$$

By a policy δ we mean a decision rule that specifies the amount $c_j = \delta(x, j)$ that we consume given that we are in state (x, j) . We let $f_\delta(x, j)$ be the expected value of U when using an admissible policy δ and starting from state (x, j) , and define $f(x, j) = \sup_\delta f_\delta(x, j)$.

A policy δ^* is said to be optimal if $f_{\delta^*} = f$. The functional equation of dynamic programming is

$$f(x, j) = \sup_{0 \leq c_j \leq x_j + D_j} (u(c_j) + \alpha E f(T(x, j))). \quad (2)$$

Some useful notation is

$$\begin{aligned} h((x, j), c, v) &= u(c) + \alpha E v(T(x, j)) \\ (Av)(x, j) &= \sup_{0 \leq c \leq x + D_j} h((x, j), c, v). \end{aligned}$$

Thus Eq. (2) can be written as

$$f(x, j) = Af(x, j), \quad (3)$$

so that the problem of finding solutions to (2) is then equivalent to the problem of finding fixed points of A .

An interpretation that can be given to be function $h((x, j), c, v)$ is that

it represents the expected return in a one-period model where the state is (x, j) , the decision is c , and v is the terminal reward function. In turn, $(Av)(x, j)$ represents the expected return in the same situation when an optimal decision is made. Often the v chosen will be the optimal return function.

Let v be fixed, and for a given state (x, j) let $c^*(x, j)$ be the (feasible) value of c which maximizes $h((x, j), c, v)$. If both v and u are concave functions, then it is known (and also very easy to prove) that both

$$c^*(x, j) \text{ and } x - c^*(x, j) \text{ are nondecreasing functions of } x. \quad (4)$$

In the event that there is more than one optimal decision we let $c^*(x, j)$ be the smallest such decision.

In this paper we will assume that we are only considering utility functions, a rate of interest, and a discount factor, such that the optimal return function f is finite-valued and can be obtained by the method of successive approximations

$$f = \lim_{n \rightarrow \infty} A^n v, \quad (5)$$

where $v \in V$ and $A^n v$ means A applied n times to v . The function space V was defined in [12] and is made up of functions v defined on the state space $R_+ \times I_+$ into R which are increasing, concave, and continuous in x for each $j \in I_+$. Furthermore v must satisfy the following boundedness condition

$$\sup_{s \in R_+, j \in I_+} \{ |v(x, j)| / \max(|u(x)|, D) \} < \infty,$$

where D is a positive constant. In order to establish (5), one needs to verify (for the given utility function, rate of interest, discount factor) that the range of $A: V$ is V and that A is a contraction mapping. We also have that f belongs to V since $A^n v$ belongs to V and V is closed [12]. In [12] this question was examined in detail for the isoelastic functions. For example, with the log utility function it was shown that a unique finite valued f satisfying (3) exists if we restrict $\epsilon(x_j + D_j) \leq c_j \leq (1 - \epsilon)(x_j + D_j)$ for any fixed $\epsilon > 0$. In order to go to the case here of $0 \leq c_j \leq (x_j + D_j)$ one needs to go through the exercise of showing the nonoptimality (with respect to f) of the newly admissible c_j . The difficulty is that the basic papers of discrete dynamic programming, Blackwell [1] and Denardo [3], require that the reward function be bounded over all admissible states and decisions, an assumption which is not satisfied by any unbounded u . Only recently have techniques been developed which

get away from this restriction [7, 9, 10]. Fortunately, there is no difficulty whatsoever in the finite period case, so our results apply without qualification for all $u \in F$.

THEOREM 1. *A concave function $g: I \rightarrow R$, where the convex set $I \subset R$, belongs to F if and only if for every set of λ_i , Δ_i , $i = 1, \dots, n$, satisfying $\lambda_i \geq 0$, $\sum \lambda_i = 1$, and $\sum \lambda_i \Delta_i = 0$,*

$$g(x_1) - \sum_{i=1}^n \lambda_i g(x_1 + \Delta_i) \geq g(x_2) - \sum_{i=1}^n \lambda_i g(x_2 + \Delta_i), \quad (6)$$

where $x_2 \geq x_1$, x_2, x_1 are in the interior of I . From Rockafeller [14, Theorems 23.1, 24.1, 24.2, Corollary 24.2.1] the convexity of g' implies that

(a) *the right-hand and left-hand derivatives of g' , g'_+ , and g'_- , exist everywhere on the interior of I , are increasing, and satisfy $g'_+ \geq g'_-$;*

(b) *for any $a, b \in I$*

$$g'(b) - g'(a) = \int_a^b g''_+(t) dt = \int_a^b g''_-(t) dt.$$

Proof. We begin the proof of the "if" part by establishing that g' exists everywhere on the interior of I . Assume the contrary, that is for some x , the derivative does not exist. Since g is concave, both the right- and left-hand derivatives exist at x and we must have $g'_+(x) - g'_-(x) = k < 0$, or

$$\lim_{y \downarrow 0} \frac{g(x+y) - g(x) - g(x) + g(x-y)}{y} = k < 0.$$

Equation (6) implies that $(-g(x) + (g(x+y)/2) + (g(x-y)/2))$ is an increasing function of x . Consequently, the derivative cannot exist for an $x' < x$ which is inconsistent with the concavity of g .

It remains to show that for any $x_2 > x_1$ in the interior of I and $0 < \lambda < 1$,

$$-g'(\lambda x_1 + (1-\lambda)x_2) + \lambda g'(x_1) + (1-\lambda)g'(x_2) \geq 0.$$

The derivatives equal the right hand derivatives so that the left hand side equals

$$\begin{aligned} \lim_{y \downarrow 0} (1/y) [-g(\lambda x_1 + (1-\lambda)x_2 + y) + g(\lambda x_1 + (1-\lambda)x_2) \\ + \lambda g(x_1 + y) - \lambda g(x_1) + (1-\lambda)g(x_2 + y) - (1-\lambda)g(x_2)]. \end{aligned}$$

The term in brackets is nonnegative, since by (6)

$$\begin{aligned} & g(\lambda x_1 + (1 - \lambda) x_2) - \lambda g(x_1) - (1 - \lambda) g(x_2) \\ & \geq g(\lambda x_1 + (1 - \lambda) x_2 + y) - \lambda g(x_1 + y) - (1 - \lambda) g(x_2 + y). \end{aligned}$$

To establish the “only if” part, we must show that $g \in F$ implies (6) which we rewrite as

$$\sum \lambda_i [g(x_1) - g(x_1 + \Delta_i) - g(x_2) + g(x_2 + \Delta_i)] \geq 0.$$

For any i ,

$$\begin{aligned} & (g(x_1) - g(x_1 + \Delta_i) - g(x_2) + g(x_2 + \Delta_i)) \\ & = - \int_{x_1}^{x_1 + \Delta_i} g'(y) dy + \int_{x_2}^{x_2 + \Delta_i} g'(y) dy, \\ & = - \int_{x_1}^{x_1 + \Delta_i} \left[g'(x_1) + \int_{x_1}^y g''(z) dz \right] dy \\ & \quad + \int_{x_2}^{x_2 + \Delta_i} \left[g'(x_2) + \int_{x_2}^y g''(z) dz \right] dy, \end{aligned}$$

using (b) of Theorem 1,

$$\geq -\Delta_i [g'(x_1) - g'(x_2)], \quad (7)$$

since g'' is increasing. To see this inequality, observe that $g''(z_2) \geq g''(z_1)$ where z_2 is the same distance above x_2 that z_1 is above x_1 when $\Delta_i > 0$. If $\Delta_i < 0$ then $y < x_1$ or $y < x_2$ as the case may be, and $g''(z_2) \geq g''(z_1)$ where z_2 is the same distance below x_2 that z_1 is below x_1 . Therefore

$$\begin{aligned} & \sum \lambda_i (g(x_1) - g(x_1 + \Delta_i) - g(x_2) + g(x_2 + \Delta_i)) \\ & \geq - \sum \lambda_i \Delta_i (g'(x_1) - g'(x_2)) = 0. \end{aligned}$$

Q.E.D.

We note that F is large enough to include utility functions u whose absolute risk aversion, $-u''/u'$, is decreasing. This is true since u' is decreasing by concavity and therefore we must have $-u''$ decreasing.

We also want the result that if $g \in F$ and Z is any random variable with zero expectation such that $Eg(x_1 + Z)$ and $Eg(x_2 + Z)$ exist, then

$$E[g(x_1) - g(x_1 + Z) - g(x_2) + g(x_2 + Z)] \geq 0, \quad \text{when } x_2 > x_1. \quad (8)$$

This follows from (6) if Z is a simple function. In order to go from simple functions to random variables we apply the same method of proof as that in Chung [2, Theorem 9.1.4]. The result is also true if we replace x_1 and x_2 by $x_1 + X$ and $x_2 + X$ where X is a random variable, and Z is a random variable such that $E(Z | X = x) = 0$ for all x . Again assuming that all expectations are defined we get (by conditioning on $X = x$)

$$E[g(x_1 + X) - g(x_1 + X + Z) - g(x_2 + X) + g(x_2 + X + Z)] \geq 0. \quad (9)$$

If X is a nonnegative random variable with a finite mean, then $v(x) = Eg(x + X)$ also exists and is concave. If we let Z be the discrete random variable $P(Z = \Delta_i) = \lambda_i$, where $\Delta_i, \lambda_i, i = 1, \dots, n$, have the properties of those same terms in (6), and be independent of X , then (9) shows that $v \in F$.

Next, we examine the idea of increasing risk as defined by Rothschild and Stiglitz [15, 17]. There they establish the equivalence of three measures of risk when comparing two random variables. The definition most useful for our purposes is that Y is more risky than X if and only if

$$Y \underset{d}{=} X + Z$$

where $\underset{d}{=}$ means "has the same distributions as" and Z is a random variable such that $E(Z | X = x) = 0$ for all x . Clearly from (9) we have that if Y is more risky than X (and all expectations exist) then $g \in F$ and $x_2 > x_1$ imply that

$$E[g(x_1 + X) - g(x_1 + Y) - g(x_2 + X) + g(x_2 + Y)] \geq 0. \quad (10)$$

The proof of Theorem 2 starts with a lemma which shows that if our utility function belongs to F , then the optimal return function f belongs to F . It is just this step that is required in a multiperiod model, but not in a two period model. As we mentioned in the introduction, Schectman [19] has independently established this lemma for the finite horizon case using a quite different method of proof.

LEMMA 1. *If $u \in F$, then $f \in F$.*

Proof. We have assumed that we are only considering problems such that (5) holds. We set v equal to the 0 function, which belongs to F , and we need to show that if $g = A^n v \in F$, then $Ag = A^{n+1} v \in F$. This is sufficient since (6) can be used to easily show that the limiting function, f , belongs to F .

It is known (for a proof in this particular case, see Miller [12]) that if g is concave then Ag is concave. We need to show that if $g(x, j) \in F$ (and hence is concave) then the concave function Ag satisfies

$$Ag(x_1, j) - \sum_{i=1}^n \lambda_i Ag(x_1 + \Delta_i, j) - Ag(x_2, j) + \sum_{i=1}^n \lambda_i Ag(x_2 + \Delta_i, j) \geq 0. \quad (11)$$

Let c_1^i , $i = 1, \dots, n$, be the optimal decision (with respect to g) for the states $x_1 + \Delta_i$ and c_2 and c_1 be the optimal decisions for the states x_2 and x_1 respectively. From (4) we know that $c_2 \geq c_1$ and $x_2 - c_2 \geq x_1 - c_1$. Let c_2^i , $i = 1, \dots, n$, be the decisions associated with the state $x_2 + \Delta_i$, and be given by $c_2^i = c_2 - c_1 + c_1^i$. They are feasible ($0 \leq c_2^i \leq x_2 + D_j$) since the c_1^i are feasible, $c_2 \geq c_1$, and $x_2 - x_1 \geq c_2 - c_1$.

We have that

$$Ag(x_1 + \Delta_i, j) = u(c_1^i) + \alpha Eg(x_1 + \Delta_i - c_1^i + Y_j, j + 1),$$

and similar equations hold for x_1 and x_2 . Since the c_2^i may not be optimal

$$Ag(x_2 + \Delta_i, j) \geq u(c_2^i) + \alpha Eg(x_2 + \Delta_i - c_2^i + Y_j, j + 1).$$

Let $\Delta_1^i = c_1^i - c_1 = c_2^i - c_2$. By (7)

$$u(c_1) - u(c_1^i) - u(c_2) + u(c_2^i) \geq -\Delta_1^i(u'(c_1) - u'(c_2)),$$

since $c_2 \geq c_1$ and $u \in F$. By the development after Eq. (9), the function $v(x) = \alpha Eg(x + Y_j, j + 1)$ belongs to F . Therefore by (7)

$$\begin{aligned} & (v(x_1 - c_1) - v(x_1 + \Delta_i - c_1^i) - v(x_2 - c_2) + v(x_2 + \Delta_i - c_2^i)) \\ & \geq -(\Delta_i - \Delta_1^i)[v'(x_1 - c_1) - v'(x_2 - c_2)], \end{aligned}$$

since $v \in F$ and $x_2 - c_2 \geq x_1 - c_1$, and $(x_1 + \Delta_i - c_1^i) - (x_1 - c_1) = (x_2 + \Delta_i - c_2^i) - (x_2 - c_2) = \Delta_i - \Delta_1^i$. Combining the above equalities and inequalities we have that the left-hand side of (11) is greater than or equal to

$$- \sum \lambda_i [\Delta_1^i(u'(c_1) - u'(c_2)) + (\Delta_i - \Delta_1^i)(v'(x_1 - c_1) - v'(x_2 - c_2))]. \quad (12)$$

If both c_1 and c_2 are interior points of their respective constraint sets, $0 \leq c_1 \leq x_1 + D_j$ and $0 \leq c_2 \leq x_2 + D_j$, then $u'(c_1) = v'(x_1 - c_1)$ and $u'(c_2) = v'(x_2 - c_2)$, by the optimality of c_1 and c_2 and the fact that the derivatives of u and v exist everywhere in the interior. In this case (12) equals $-\sum \lambda_i \Delta_i [u'(c_1) - u'(c_2)] = 0$.

We will consider the boundary cases of $c_1 = 0$ or $c_1 = x_1 + D_j$ and $c_2 = 0$ or $c_2 = x_2 + D_j$ by giving the proofs for the cases $c_1 = 0$ and $c_2 = 0$ only. A similar situation arises in the proof of Theorem 2, and there we give the proofs of the cases $c_1 = x_1 + D_j$ and $c_2 = x_2 + D_j$ only.

One possibility at the boundary is c_1 at a boundary, say $c_1 = 0$, but c_2 is not. In order to apply (12) where c_1 is a left end point we need to verify that

$$u(x) - u(x + \Delta_i) \geq - \int_x^{x+\Delta_i} \left[u'_+(x) + \int_x^y u''_+(z) \right] dz \quad (13)$$

for $x = c_1$. Since u is nondecreasing $u(c_1) \leq \lim_{x \downarrow c_1} u(x)$. If $u(c_1) < \lim_{x \downarrow c_1} u(x)$, then $u'_+(c_1) = +\infty$ and (13) holds. If $u(c_1) = \lim_{x \downarrow c_1} u(x)$, then (13) holds for $x = c_1$ since $u'_+(c_1) \geq \lim_{x \downarrow c_1} u'_+(x) = \lim_{x \downarrow c_1} u''_+(x)$ [14, Theorem 24.1], and (13) holds with equality for all $x > c_1$. Returning to the main argument, the optimality of c_1 implies that $u'_+(c_1) \leq v'(x_1 - c_1)$ and clearly $\Delta_1^i \geq 0$ for all i . Therefore (12) is greater than or equal to

$$- \sum \lambda_i [\Delta_1^i v'(x_1 - c_1) + (\Delta_i - \Delta_1^i) v'(x_1 - c_1)] - \sum \lambda_i \Delta_i (-u'(c_2)) = 0.$$

The other possibility at the boundary is c_2 at a boundary, say $c_2 = 0$. By (4) $c_1 = 0$, and therefore $c_1^i = c_2^i$. Then (11) becomes

$$v(x_1) - \sum \lambda_i v(x_1 + \Delta_i - c_1^i) - v(x_2) + \sum \lambda_i v(x_2 + \Delta_i - c_2^i).$$

This quantity is nonnegative since $v(x_1) - \sum \lambda_i v(x_1 + \Delta_i) - v(x_2) + \sum \lambda_i v(x_2 + \Delta_i) \geq 0$ by (6) since $v \in F$, and $v(x_1 + \Delta_i) - v(x_1 + \Delta_i - c_1^i) - v(x_2 + \Delta_i) + v(x_2 + \Delta_i - c_2^i) \geq 0$ for all i since v is concave.

THEOREM 2. *Let $u \in F$ and $X_1, X_2, \dots$ be a sequence of random variables describing labor income (Case a), and $Y_1, Y_2, \dots$ be a second sequence of random variables describing labor income (Case b). If for each i , Y_i is riskier than X_i , then the optimal amount to consume as a function of the state (x, j) in Case a is greater than the optimal amount to consume in Case b.*

Verifying the hypothesis of the following lemma leads directly to a proof of Theorem 2.

LEMMA 2. *Let f_X be the optimal return function in Case a and f_Y be the optimal return function in Case b. If $d(x, j) = f_Y(x, j) - f_X(x, j)$ is a nondecreasing function of x then the conclusion of Theorem 2 holds.*

Proof. By Lemma 1 we know that $f_X, f_Y \in F$. Let c^* be the optimal decision for state (x, j) with the optimal return function f_X . For $c > c^*$,

$$\begin{aligned} & h((x, j), c^*, f_Y) - h((x, j), c, f_Y) \\ &= u(c^*) + \alpha E f_X(r(x - c^*) + Y_j, j + 1) \\ & \quad + \alpha E d(r(x - c^*) + Y_j, j + 1) \\ & \quad - u(c) - \alpha E f_X(r(x - c) + Y_j, j + 1) \\ & \quad - \alpha E d(r(x - c) + Y_j, j + 1). \end{aligned}$$

Since d is nondecreasing and $c > c^*$ we have that the right-hand is

$$\begin{aligned} & \geq u(c^*) + \alpha E f_X(r(x - c^*) + Y_j, j + 1) \\ & \quad - u(c) - \alpha E f_X(r(x - c) + Y_j, j + 1) \\ & \geq u(c^*) + \alpha E f_X(r(x - c^*) + X_j, j + 1) \\ & \quad - u(c) - \alpha E f_X(r(x - c) + X_j, j + 1) \end{aligned}$$

by (10) since $r(x - c^*) > r(x - c)$ and $f_X \in F$

$$= h((x, j), c^*, f_X) - h((x, j), c, f_X) \geq 0, \text{ since } c^* \text{ is optimal.}$$

Therefore the optimal amount to consume in Case b is less than or equal to c^* . Recall that in case of ties (which could not happen with strict concavity) we pick the smallest c . Q.E.D.

The following lemma from [12] is needed to establish that d is nondecreasing. Like f , h and A vary with Case a and Case b, but we suppress this dependence in the notation.

LEMMA 3. *Consider the model in the case where $Y_1, Y_2, \dots$ are the random variables describing labor income (Case b). Let $v \in V$ and suppose that v satisfies Condition A below. Then $f_Y(x, j) - v(x, j)$ is a nondecreasing function of x .*

CONDITION A. *Given any two states x_1, j and x_2, j , $x_2 > x_1$, and decision c_1 for (x_1, j) , there is a feasible decision c_2 for (x_2, j) such that*

- (a) $x_2 - c_2 \geq x_1 - c_1$,
- (b) $(v(x_1, j) - h((x_1, j), c_1, v) - v(x_2, j) + h((x_2, j), c_2, v)) \geq 0$.

Equation (b) by itself is a necessary and sufficient condition that $Av - v$ be a nondecreasing function. The proof of the lemma consists of verifying an induction hypothesis in order to show that $A^n v - v$ is nondecreasing, and using (5) $f_Y = \lim_{n \rightarrow \infty} A^n v$.

Proof of Theorem 2. By Lemma 2 and Lemma 3 we need to show that Condition A holds where we let $v = f_X$. Recall that below Eq. (5) we showed that f belongs to V . Let c_1^* and c_2^* be the optimal decisions for states (x_1, j) and (x_2, j) with the optimal return function f_X . Given a c_1 we set $c_2 = c_1 + c_2^* - c_1^*$.

Since c_1^* and c_2^* satisfy (6), $c_2^* \geq c_1^*$, and $x_2 - x_1 \geq c_2^* - c_1^*$. Thus c_2 is feasible since c_1 is feasible, and $c_2 - c_1 = c_2^* - c_1^* \leq x_2 - x_1$ and (a) of Condition A holds.

Recalling that $f_X = Af_X$, the left-hand side of (b) in Lemma 3 equals

$$\begin{aligned} & u(c_1^*) + \alpha Ef_X(x_1 - c_1^* + X_j, j + 1) - u(c_1) - \alpha Ef_X(x_1 - c_1 + Y_j, j + 1) \\ & \quad - u(c_2^*) - \alpha Ef_X(x_2 - c_2^* + X_j, j + 1) \\ & \quad + u(c_2) + \alpha Ef_X(x_2 - c_2 + Y_j, j + 1) \\ & \geq u(c_1^*) + \alpha Ef_X(x_1 - c_1^* + X_j, j + 1) \\ & \quad - u(c_1) - \alpha Ef_X(x_1 - c_1 + X_j, j + 1) \\ & \quad - u(c_2^*) - \alpha Ef_X(x_2 - c_2^* + X_j, j + 1) \\ & \quad + u(c_2) + Ef_X(x_2 - c_2 + X_j, j + 1) \end{aligned} \quad (14)$$

by (10) since $x_2 - c_2 \geq x_1 - c_1$ and $f_X \in F$.

Now let $\Delta = c_2 - c_2^* = c_1 - c_1^*$, and we will show that (14) is non-negative. By (7) $(u(c_1^*) - u(c_1) - u(c_2^*) + u(c_2)) \geq -\Delta(u'(c_1^*) - u'(c_2^*))$, since $u \in F$ and $c_2^* \geq c_1^*$. As in the proof of Lemma 1 we let $v(x) = \alpha Eg(x + X_j, j)$. Then $v \in F$ and $(v(x_1 - c_1^*) - v(x_1 - c_1) - v(x_2 - c_2^*) + v(x_2 - c_2)) \geq \Delta(v'(x_1 - c_1^*) - v'(x_2 - c_2^*))$. Therefore (14) is greater than or equal to

$$\Delta(-u'(c_1^*) + v'(x_1 - c_1^*) + u'(c_2^*) - v'(x_2 - c_2^*)). \quad (15)$$

If both c_1^* and c_2^* are interior points of their constraints sets, then $u'(c_1^*) = v'(x_1 - c_1^*)$, $u'(c_2^*) = v'(x_2 - c_2^*)$ and (15) equals zero. If c_1^* is at a boundary, say $c_1^* = x_1 + D_j$, and c_2^* is not, then $u'(c_1^*) \geq v'_+(x_1 - c_1^*)$ and $u'(c_2^*) = v'(x_2 - c_2^*)$. Since Δ must be nonpositive in this case, (15) will be nonnegative. Here we use (13) as applied to v . If c_2^* is at a boundary, say $c_2^* = x_2 + D_j$, then c_1^* must equal $x_1 + D_j$, and $v(x_2 - c_2^*) = v(x_1 - c_1^*)$ and $v(x_2 - c_2) = v(x_1 - c_1)$. Then (14) becomes $(u(c_1^*) - u(c_1) - u(c_2^*) + u(c_2))$ which is nonnegative by the concavity of u . Q.E.D.

It is also interesting to verify the plausible result that a less risky income stream is more desirable, that $f_X \geq f_Y$. Let A_X and A_Y refer to Case a

and Case b respectively. If v is a function which satisfies $A_Y v \leq v$ then by monotonicity $A_Y^2 v \leq A_Y v$, and hence $A_Y^n v \leq v$ which shows that $f_Y = \lim_{n \rightarrow \infty} A_Y^n v \leq v$. We let $v = f_X$ and the argument is complete when we show that $A_Y f_X \leq f_X = A_X f_X$. For any feasible decision c , $u(c) + \alpha E f_X(x - c + Y_j, j + 1) \leq u(c) + \alpha E f_X(x - c + X_j, j + 1)$ since f_X is concave and Y_j is more risky than X_j . This inequality is in fact an equivalent definition of riskiness [15]. Maximizing over c gives the desired result.

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